XCREATIVS TECHNOLOGIES · CONFIDENTIAL · INTERNAL

Gigmann Executive Cockpit

Proof of Concept — Product & Engineering Specification

Working codename: *"Ahenfie"* — the seat from which the chief sees and governs everything. (Internal codename for this build; rename freely.)

> **THE ONE LINE**
>
> We are not building a dashboard. We are building a system that thinks about Gigmann's entire network while Sammy sleeps, and tells him — the moment he opens his phone — the three things that need him today. The cockpit is the proof; the goal is that he commits to build the rest.

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Contents

1. The one move **3**
2. The hero moment — "It briefed me" **3**
2.1 The experience 3
2.2 What the brief sounds like (illustrative, from the synthetic network) 4
2.3 Why it converts 4
2.4 The follow-through (the second wow) 4
3. Product scope & demo narrative **5**
3.1 In scope (this PoC) 5
3.2 Explicitly out of scope (presented as the roadmap) 5
3.3 The demo narrative (how we present it) 5
4. The synthetic Gigmann network **6**
4.1 The twelve facilities (summary) 6
4.2 Ghana realities the data must carry 6
5. Feature specification — the cockpit **7**
5.1 The Daily Brief (hero) 7
5.2 The Network (single pane of glass) 7
5.3 Facility detail (drill-down) 8
5.4 Executive KPIs 8
5.5 The Attention Feed 8
5.6 Ask (natural-language query + generated actions) 8
5.7 My Day (personal task & workflow system) 8
5.8 Approvals & decision routing 9
5.9 Delegation & follow-through 9
5.10 Reports 9
5.11 Notifications & alerts 9
5.12 Personalisation 9
6. The intelligence design **9**
6.1 The governing principle 9
6.2 The Daily Brief pipeline 10
6.3 The signal engine (computed, not AI) 10
6.4 Natural-language query 10
6.5 Generated actions 10
6.6 Model, prompts, cost, fallback 11
7. Data model **11**
8. Technical architecture & stack **12**
8.1 Recommended stack 12
8.2 System shape 12
8.3 Production note 13
8.4 Security for the PoC 13
9. UX & screens **13**

9.1 Design language 13
9.2 Screen inventory 13
9.3 Mobile-first, desktop-strong 13
9.4 The “alive” details 13
10. Synthetic data generation strategy 14
11. Build plan (sprint) 14
11.1 Workstreams 14
11.2 Demo-readiness gate 15
12. Definition of done 15
Appendix A — Facility roster (detail) 16
Appendix B — KPI definitions 16
Appendix C — Planted stories (demo material) 17

1. The one move

This proof of concept exists to do one job: make Sammy Adjei, CEO of Gigmann Medicals, *feel* something he has never felt from a software demo, and commit money to build the full platform on the strength of that feeling. Everything in this spec is bent toward that single outcome.

Sammy runs a fast-growing network of hospitals and diagnostic centres across Ghana. He has told us his world: facilities that can't share data, an ERP that only stores, dashboards he opens side by side to compare, and no single place from which to run it all. Of the three pillars we identified with him — interoperability, an intelligence layer, and an executive cockpit — **the cockpit is the one he can feel in sixty seconds**, because it is where he personally lives every day. So we build the cockpit first, make it stunning, and present the other two as the roadmap it unlocks.

The category shift we are selling is simple: not a dashboard he operates, but **a chief of staff that works while he sleeps**. A passive tool shows numbers when asked. Ahenfie watches the whole network continuously and, the moment he opens his phone, has already written his brief — the three things that need him today, the worst one first, in plain English. That is the intrigue, and it is what makes this worth paying for.

> **DECISIONS LOCKED FOR THIS BUILD**
>
> **Hero moment:** the AI-written "Daily Brief" — the cockpit briefs him before he asks.
>
> **Real AI:** live Claude on the magic touchpoints (the brief, the natural-language ask, generated actions). Mock data underneath, real intelligence on top.
>
> **Data:** an invented but deeply Ghana-grounded network of 12 facilities — real towns, NHIS, mobile money, real disease and payer patterns. We do not use or fake Sammy's real data.
>
> **Build quality:** a fully functional product, not a clickable mock. Real backend, real database, real AI service.
>
> **Surfaces:** mobile-first (shown on his phone), with a first-class desktop view.

2. The hero moment — "It briefed me"

Everything else in the cockpit is table stakes. **This is the moment that closes the deal.** Build it first, build it best, and protect its quality above all other features.

2.1 The experience

Sammy opens Ahenfie on his phone in the morning. Before he taps anything, before he searches for a single number, the top of the screen already holds a short, sharp brief of his entire network — dated this morning, written in plain language, as if a brilliant chief of staff had stayed up reviewing all twelve facilities and left a note on his desk. It opens with a greeting, names the worst problem first, connects cause to effect, points to the one bright spot worth copying, lists what is waiting for his sign-off, and reassures him that everything else is steady.

The emotional beat we are engineering is the realisation: *"this thing was working while I slept, and it already knows what I need to do."* That feeling — of being looked after by an intelligence that never sleeps — is the product.

2.2 What the brief sounds like (illustrative, from the synthetic network)

> *Good morning, Sammy. Tuesday, 9 June. Here's your network.*
>
> 🔴 **Tafo Maternity needs you first.** Revenue is down 22% over two weeks — but demand hasn't fallen; deliveries and OPD visits are flat. The drop tracks almost exactly with a pile-up of NHIS claims that were never submitted. This looks like a claims-officer breakdown, not a patient problem. About GH₵ 78,000 is sitting unbilled. [Why →] [Message the manager →]
>
> 🟠 **Asokwa will run dry.** At current usage, Asokwa Diagnostic runs out of malaria RDT kits in about 5 days. The supplier's lead time is 7. If we don't reorder today, testing stops by the weekend. [Approve reorder →]
>
> 🟢 **Adansi is your bright spot.** Best week this quarter — OPD up 14%, claims clean, no stock issues. Worth learning what the manager is doing right and copying it to Kasoa and Nima. [See Adansi →]
>
> Two things are waiting for your sign-off: a GH₵ 85,000 ultrasound for Assin Fosu, and a new Medical Officer for Kasoa.
>
> Everything else is steady. Network revenue is tracking GH₵ 5.8M this month. NHIS owes you GH₵ 1.9M — up GH₵ 140k this week, worth watching. All 12 facilities are open and staffed.

2.3 Why it converts

- **It is proactive, not passive.** It tells him what matters before he asks. Every other vendor shows him a screen and waits. This one did the thinking.
- **It prioritises.** An overwhelmed executive's scarcest resource is attention. Surfacing the single worst thing first, by impact, is the most valuable thing software can do for him.
- **It diagnoses, not just reports.** "Revenue is down" is a number. "Revenue is down because claims aren't being submitted" is intelligence. That causal leap is the moment he believes the system is genuinely smart.
- **It speaks his language.** Conversational, in cedis, about NHIS and RDT kits and claims officers — his world, not generic SaaS.
- **It is genuinely alive.** Because it is real AI on live network state, when he asks "can it do X?" the honest answer is usually yes. That is impossible to fake convincingly, and it is our advantage.

2.4 The follow-through (the second wow)

Directly beneath the brief, he can act. Tapping **Why →** on the Tafo item makes the system dig deeper, live, and explain. Tapping **Message the manager →** has it draft a short, firm WhatsApp-style message to the Tafo manager that he can send. And a single input lets him ask his business anything in plain English — *"which facility needs me this week?"*, *"summarise Kasoa for* *the board"* — and get a grounded answer or a generated document. The cockpit does not just show him work; it does work.

> **DESIGN MANDATE FOR THE HERO**
>
> The brief must feel: alive (it changes with the data and the day), personal (it speaks to him by name, in his idiom), smart (it connects dots a dashboard never would), and fast (it appears within a moment of opening). If any of those four slips, the magic dies. Treat brief quality as the project's top acceptance criterion.

3. Product scope & demo narrative

3.1 In scope (this PoC)

The full executive cockpit: the Daily Brief, a single-pane network view of all 12 facilities, facility drill-down, executive KPIs, an attention feed, natural-language query with generated actions, a personal task/workflow system, approvals and decision routing, delegation, and live report generation. All fully functional, mobile-first with a strong desktop view, and powered by real Claude on the intelligent surfaces.

3.2 Explicitly out of scope (presented as the roadmap)

- **The interoperability engine itself.** Real FHIR exchange, the master patient index, and live ERPNext adapters are the post-sale build. In the PoC, cross-facility data simply exists in our database — we demonstrate the value of unified data without building the pipes yet.
- **Clinical AI.** Anything touching diagnosis or treatment is regulated and out. The cockpit's intelligence is strictly financial, operational, and managerial — which is exactly what an owner needs and where we can move fast and safely.
- **Production hardening.** Real multi-tenant security, full audit, and Ghana-hosting are production concerns. The PoC is functional and real, but built for a compelling demo, not for live patient traffic.

These are not weaknesses to hide — they are the **reason for the next conversation**. The cockpit proves the vision; interoperability and the full intelligence layer are what he pays to unlock.

3.3 The demo narrative (how we present it)

1. **Open cold.** Sammy (or we, as Sammy) open the app on a phone. The brief is already there. We say nothing for a beat and let him read it.
2. **Feel the diagnosis.** We tap "Why?" on the Tafo revenue item and let the system explain the claims breakdown live.
3. **Take an action.** We approve the Asokwa reorder, and draft the message to the Tafo manager — the cockpit does work, not just display.
4. **Command the network.** We open the network view — all 12 facilities, alive, colour-coded — and drill into one.

5. **Ask anything.** We type a plain-English question and the system answers from his data. This is the close.
6. **Then the roadmap.** We zoom out: “this is the cockpit. Underneath it sits the interoperability layer and the full intelligence engine. Here is what we build together next.”

4. The synthetic Gigmann network

We invent a believable Gigmann — twelve facilities across Ghana, grounded in real towns, real payer realities, and real clinical patterns. The data is fictional but must **feel unmistakably Ghanaian**, because that realism is what makes the demo land. Full roster in Appendix A; planted demo stories in Appendix C.

4.1 The twelve facilities (summary)

Facility	Region	Type & focus	Beds	Patients/mo	Rev GH₵/mo
Assin Fosu Specialist Hospital (flagship)	Central	Full-service specialist + pharmacy + diagnostics	60	7,800	890k
Asokwa Diagnostic & Specialist Centre	Ashanti	Diagnostics-led, imaging & labs	24	5,200	560k
Kasoa Polyclinic	Central	High-volume peri-urban OPD	40	9,400	640k
Adansi Community Hospital	Ashanti	Rural community hospital	30	4,100	380k
Takoradi Harbour Clinic	Western	Occupational & general	20	3,600	410k
Tafo Maternity & Child Health	Ashanti	Maternity & child health	25	3,900	350k
Nima Urban Health Centre	Greater Accra	Dense urban, high footfall	18	8,700	620k
Ho Central Medical Centre	Volta	Regional general	35	4,800	470k
Tamale North Clinic	Northern	General; northern disease profile	22	4,300	360k
Cape Coast Castle Clinic	Central	General + surgical theatre	28	4,600	440k
Sunyani Diagnostic Hub	Bono	Diagnostics (newly acquired, ramping)	16	2,400	180k
Sekondi Pharmacy & Clinic	Western	Retail pharmacy + clinic	12	3,100	300k

Network totals: roughly **65,000+ patients/month** and **~GH₵ 5.8M monthly revenue** across the twelve.

4.2 Ghana realities the data must carry

- **NHIS dominance.** The National Health Insurance Scheme is the largest payer at most facilities. The data must model claims submitted, paid, denied, denial rates, and — critically — slow reimbursement (large outstanding balances the Scheme owes). This is a genuine, felt pain and prime material for the brief.

- **Payer mix.** Each facility carries a realistic split of NHIS / cash & mobile money / private insurance. Urban and retail-heavy sites lean more cash and MoMo; rural and maternity sites lean heavily NHIS.
- **Mobile money.** A real and growing share of cash collection is via MTN Mobile Money. Model it as its own collection channel.
- **Clinical & seasonal pattern.** Common presentations: malaria, typhoid, hypertension, diabetes, respiratory infections, maternal care, road-traffic injuries. Malaria and related testing peak in the rainy season, which should visibly drive volume and stock burn at the right times of year.
- **Staff roles.** Use Ghanaian healthcare roles — Medical Officer, Physician Assistant, Nurse, Midwife, Lab Technician, Pharmacist, Records Officer, and NHIS Claims Officer — with licences that expire and must be tracked.
- **Operating realities.** Power instability ("dumsor") makes generators and oxygen continuity real line items; referrals flow up to teaching hospitals (Korle Bu, KATH, Cape Coast Teaching Hospital). These details make the world believable.

5. Feature specification — the cockpit

Twelve capability areas. Each lists its purpose and the functionality the team must ship. The Daily Brief (5.1) is the hero and the priority; the rest complete the command experience around it.

5.1 The Daily Brief (hero)

Purpose: greet Sammy and hand him the three things that need him today, before he asks.

- **Auto-generated each morning** (and refreshable on demand) by the intelligence service from the live network state.
- **Prioritised by impact** — worst first, with a clear severity signal (critical / watch / good).
- **Diagnostic** — connects cause and effect where the data supports it (e.g., revenue drop ↔ unsubmitted claims).
- **Actionable inline** — each item carries actions: explain further, message a manager, approve, or open the facility.
- **Honest** — figures are pulled from the database, never invented; the AI narrates and prioritises, it does not fabricate numbers.

5.2 The Network (single pane of glass)

Purpose: the whole empire on one screen, alive.

- **All 12 facilities as living tiles** — each showing its name, region, a status colour (AI-assessed health), and one or two headline numbers (today's revenue, occupancy).
- **A network pulse** — a single composite indicator at the top answering "how is my whole network right now?" at a glance.
- **Sort & filter** — by status, region, revenue, or attention needed; problems float to the top.

- **Map view (optional)** — facilities pinned on a map of Ghana, colour-coded, for a striking overview.

5.3 Facility detail (drill-down)

Purpose: one facility, in depth, one tap away.

- **KPI trends** — revenue, patients, occupancy, claims, stock, over time, with week-over-week movement.
- **Facility-level alerts and AI notes** — what the intelligence layer has flagged here.
- **Staff snapshot** — headcount by role, licence-expiry warnings.
- **Quick actions** — message the manager, create a task, open an approval, generate a facility summary.

5.4 Executive KPIs

Purpose: his numbers, portfolio-wide, the way an owner thinks.

- **Headline metrics** — network revenue, patients seen, occupancy, NHIS outstanding, unbilled (revenue leakage), payer mix, per-facility margin. Full definitions in Appendix B.
- **Comparison** — facilities ranked and compared automatically, killing the "open dashboards side by side" problem he described.
- **Drill-through** — every network number expands to its per-facility contributors.

5.5 The Attention Feed

Purpose: a running, prioritised list of everything that needs him — the brief, persisted and extended.

- **Exceptions, risks, and anomalies** surfaced by the signal engine and ranked by the AI.
- **Each item is dismissable or actionable** ; resolved items drop off, new ones surface.

5.6 Ask (natural-language query + generated actions)

Purpose: let him interrogate and command his business in plain English.

- **Ask anything** — "which facility needs me this week?", "how is Kasoa's NHIS doing?" — answered, grounded strictly in the data.
- **Generate documents** — "summarise the network for my board," "draft a message to the Tafo manager" — produced on demand.
- **Grounded and guard-railed** — answers come only from the database; if the data can't support an answer, the system says so rather than inventing.

5.7 My Day (personal task & workflow system)

Purpose: the personal operating system he asked for — his version of Notion, but wired to his facilities.

- **Tasks tied to facilities and decisions** — created manually or spun directly off a brief item or alert.
- **Priorities, due dates, status** — a clean, fast personal board.

- **"Turn this into a task"** — one tap from any insight to a tracked action.

5.8 Approvals & decision routing

Purpose: the governance layer, digitised — the things only he can authorise flow to him and he signs off from one place.

- **Approval queue** — capital purchases above a threshold, key hires, large reorders, routed to him with the context to decide.
- **Approve, decline, or ask a question** — from his phone, with the decision logged.

5.9 Delegation & follow-through

Purpose: he assigns, the system tracks, nothing falls through.

- **Assign actions to facility managers** and see completion status.
- **Follow-ups surface** if an assigned action stalls.

5.10 Reports

Purpose: investor and board reporting generated from live data, not assembled by hand.

- **One-tap network report** — performance, highlights, risks, per-facility — generated and exportable.
- **Per-investor / per-facility cuts** — framed for the audience, feeding his PE and financeability story.

5.11 Notifications & alerts

Purpose: pull him in when something genuinely needs him.

- **Push notifications** for critical items (stock-out imminent, sharp revenue drop, approval waiting).
- **Quiet by default** — only the things that matter, so notifications stay trusted.

5.12 Personalisation

Purpose: the cockpit learns what he cares about.

- **Tunable priorities and thresholds** — which metrics and which facilities he wants watched most closely (simulated learning in the PoC).

6. The intelligence design

6.1 The governing principle

> **Deterministic detection, AI narration.** Numbers, thresholds, deltas, and projections are computed in code from the database — never by the model. Claude's job is to interpret those computed signals, prioritise them, connect them, and write them in human language. The AI never invents a figure. This is what makes the intelligence both stunning and trustworthy.

6.2 The Daily Brief pipeline

7. **Assemble state.** Gather the current snapshot of all 12 facilities: latest KPIs, week-over-week deltas, open approvals, active alerts.
8. **Compute signals.** The signal engine (§6.3) flags anomalies deterministically — revenue swings, stock-out projections, claim backlogs, attrition risk — each with magnitude and supporting figures.
9. **Compose context.** The flagged signals plus relevant facility facts are packaged into a structured context object.
10. **Generate.** Claude, prompted as Sammy's chief of staff, selects the top items by impact, connects causes where the data supports it, and returns structured JSON (per item: severity, facility, headline, explanation, suggested actions) plus a short prose brief.
11. **Render & cache.** The cockpit renders the structured brief with inline actions; the result is cached for the day and regenerates on demand or on material change.

6.3 The signal engine (computed, not AI)

- **Trend & delta detection** — week-over-week and trailing-window movement on revenue, volume, occupancy, claims; flags swings beyond a threshold.
- **Stock-out projection** — stock level ÷ daily burn vs supplier lead time; flags items that will run out inside the reorder window.
- **Claims health** — submission gaps (revenue recorded but claims not submitted), denial-rate spikes, growing outstanding reimbursement.
- **Revenue leakage** — services delivered but unbilled.
- **Staff signals** — licence expiries approaching, attrition-risk indicators, deployment imbalances.

Each signal carries its own numbers. The AI receives *facts*, and turns them into a prioritised, readable, human brief.

6.4 Natural-language query

12. **Interpret the question** and identify the facilities, metrics, and timeframe involved.
13. **Retrieve** the relevant data via structured queries (with pgvector for fuzzy matching over facility notes and names).
14. **Answer, grounded**, in plain language from the retrieved data — and, where asked, produce an artifact (a drafted message, a mini-report).

- **Guardrail** the model answers only from supplied data; if the data can't support an answer, it says so. No fabrication.

6.5 Generated actions

Beyond answering, the AI produces work: a firm but professional message to a facility manager, a board-ready facility summary, a draft of the network report. Each is editable before it leaves his hands.

6.6 Model, prompts, cost, fallback

- **Model** Claude Sonnet for the brief, query, and generation — the balance of quality and speed the live demo needs; structured JSON outputs for the brief and insights.
- **Prompt architecture** a stable system role ("you are Sammy's chief of staff…") + the structured context per call; strict instructions to use only supplied figures.
- **Cost & latency** cache the daily brief; cache repeated queries; pre-warm the morning brief so it is instant on open.
- **Fallback** if the API is unavailable mid-demo, serve the last cached brief and degrade gracefully — the cockpit never shows a broken screen.

7. Data model

A relational core in PostgreSQL with time-series metrics. Key entities below; field lists are indicative, not exhaustive.

facilities

id, name, region, town, type, beds, opened_date, status (active / ramping / flagship), manager_name, payer_mix (nhis / cash_momo / private), latitude, longitude.

facility_metrics (time-series — daily/weekly)

facility_id, date, revenue, cash_revenue, momo_revenue, patients_seen, admissions, occupancy_rate, avg_wait_minutes, nhis_claims_submitted, nhis_claims_paid, nhis_claims_denied, nhis_outstanding, unbilled_amount.

inventory_items

id, facility_id, item_name, category, stock_level, daily_burn, reorder_point, lead_time_days, unit_cost.

staff

id, facility_id, name, role, licence_number, licence_expiry, status, attrition_risk, joined_date.

alerts

id, facility_id, type, severity, title, detail, supporting_figures (json), status, created_at.

tasks

id, title, detail, facility_id (nullable), priority, status, due_date, assigned_to, created_by, source (manual / brief / alert), created_at.

approvals

id, type (capital / hire / reorder), facility_id, amount, title, context, requested_by, status, decided_at, decision_note, created_at.

briefs / insights

briefs: id, date, prose, items (json), generated_at, model, source_signal_ids. insights: id, type, facility_id, content, supporting_figures (json), generated_at.

users

id, name, role (executive / facility_manager), facility_id (for managers), preferences (json: watched metrics, thresholds).

8. Technical architecture & stack

Chosen for a fully functional, AI-native, real-time, mobile-first product with a first-class desktop view. Aligned to the XCreativs house stack where it serves the goal, upgraded where the demo demands.

8.1 Recommended stack

Layer	Choice	Why
Frontend	Next.js (App Router) + React + TypeScript	House standard; SSR + PWA for an installable, app-like mobile experience and a strong desktop view from one codebase.
UI & styling	Tailwind CSS + shadcn/ui + Tremor	Fast, consistent, premium components; Tremor is purpose-built for dashboards and KPI tiles.
Motion	Framer Motion	The “alive” feel — the subtle motion that makes the cockpit feel like it’s thinking.
Data fetching	TanStack Query	Caching, background refresh, and the live feel without hand-rolled state.
Backend	NestJS (TypeScript)	House standard; structured, modular API and a clean home for the intelligence service.
Database	PostgreSQL + TimescaleDB + pgvector	Relational core; TimescaleDB for facility metric time-series; pgvector for grounded natural-language retrieval.
Cache / realtime	Redis + WebSockets (Socket.io)	Cache AI briefs and queries; push live updates so the cockpit feels continuously awake.
AI	Anthropic Claude (Sonnet) via an intelligence service	The brief, the query, and generated actions — real intelligence, structured JSON outputs, grounded in the DB.
Auth	Auth.js (NextAuth)	Real but lightweight sign-in for the demo (Sammy as executive; a manager persona).
Infra (PoC)	Vercel (web) + managed Postgres (Neon) + container host (Railway/Fly)	Fast to ship and demo. Production note below.
Data seeding	Node generator service	Produces the Ghana-grounded time-series with deterministic planted stories (§10).

8.2 System shape

- **Web app (Next.js)** renders the cockpit, mobile-first, installable as a PWA, responsive to a full desktop layout.
- **API (NestJS)** serves facilities, metrics, tasks, approvals, alerts, and brokers all AI calls.
- **Intelligence service** (inside or beside the API) runs the signal engine, assembles context, calls Claude, returns structured briefs, answers, and generated documents.

- **Data layer** Postgres/TimescaleDB for state and history; Redis for caching and realtime; pgvector for retrieval.
- **Realtime channel** pushes new alerts and brief updates to the open cockpit.

8.3 Production note

For the PoC, cloud hosting is correct — speed wins. For the eventual production system, Gigmann's positioning and Ghana's data realities point to **Ghana-based hosting** and full data-protection compliance. Architect cleanly now so that move is a deployment decision later, not a rebuild.

8.4 Security for the PoC

- **Real auth, fake data** — genuine sign-in, but no real patient or facility data anywhere in the system. The synthetic network protects everyone and lets us control the demo.
- **No clinical detail** — the cockpit deals in operational and financial signals, not patient clinical records.

9. UX & screens

9.1 Design language

Premium, calm, and fast. Dark-mode-capable. The cockpit should feel like a **command instrument** — confident typography, generous space, restrained colour used to signal status (critical / watch / good), and just enough motion to feel alive without fuss. It is the seat of someone who runs an empire; it should feel that way in his hand.

9.2 Screen inventory

- **Home / The Brief** — the hero. The morning brief, the attention feed, approvals waiting. The first and most important screen.
- **Network** — all 12 facilities as living tiles, the network pulse, sort/filter, optional Ghana map.
- **Facility detail** — KPI trends, alerts, staff snapshot, quick actions.
- **Ask** — the natural-language query surface and generated-document output.
- **My Day** — the personal task and workflow board.
- **Approvals** — the decision queue.
- **Reports** — generate and export network/investor/board reports.

9.3 Mobile-first, desktop-strong

- **Mobile (primary)** designed to be shown on a phone in his hand: the brief leads, thumb-reachable actions, a bottom nav across the core screens.
- **Desktop (first-class)** the same product in a multi-pane command layout — network and detail side by side, denser KPI grids — quality, not an afterthought.

9.4 The "alive" details

- **Subtle motion** as the brief composes and numbers settle — it should feel like the system is thinking, then speaking.
- **Live updates** tiles and the pulse shift as data changes, reinforcing that the cockpit is always awake.

10. Synthetic data generation strategy

The data must be realistic enough to be believed and scripted enough to guarantee the demo lands every time.

- **Realistic baselines** — each facility gets plausible ranges for revenue, volume, occupancy, and payer mix consistent with its type, size, and region (Appendix A).
- **Time-series with texture** — generate daily/weekly history with natural variation, weekday/weekend patterns, and rainy-season malaria peaks driving volume and stock burn.
- **Deterministic planted stories** — inject the specific narratives the brief will surface (Appendix C) so the demo always tells the same compelling story: Tafo's claims breakdown, Asokwa's impending stock-out, Adansi's star week, Kasoa's denial spike, and the rest.
- **Grounded specifics** — cedis, NHIS, MoMo, RDT kits, Ghanaian names and roles throughout. No generic placeholder data anywhere.
- **Reseedable & stable** — a single seed regenerates the whole network identically, so the demo state is reproducible on demand.

11. Build plan (sprint)

A focused sprint. Sequence is deliberate: data and the AI brief first (the hero), then the surfaces around it, then polish. Adjust durations to team size; the order is what matters.

Phase	Focus	Output / milestone
Phase 0 — Foundations	Repo, stack, schema, auth, CI; the Ghana-grounded data generator and seed.	A seeded database of the 12-facility network with realistic history and planted stories.
Phase 1 — The hero	Signal engine + intelligence service + Claude integration; the Daily Brief end to end.	The morning brief generates live from real signals and renders with inline actions. The magic works.
Phase 2 — Command surfaces	Network view + pulse, facility detail, executive KPIs, attention feed.	Sammy can see and drill the whole network; problems surface to the top.
Phase 3 — Act & ask	Ask (NL query + generated actions), My Day, Approvals, delegation, reports.	The cockpit does work and answers questions, not just displays.
Phase 4 — Polish & demo	Motion, mobile + desktop refinement, performance, caching, fallback, demo run-throughs.	Demo-ready: fast, alive, flawless on the demo path, on phone and laptop.

11.1 Workstreams

- **Data & backend** schema, generator, API, signal engine.

- **Intelligence** Claude integration, prompt design, structured outputs, grounding, caching.
- **Frontend** the cockpit surfaces, mobile-first + desktop, the alive feel.
- **Design** the command-instrument language, the brief, the network view.

11.2 Demo-readiness gate

Nothing ships to the meeting until the demo narrative (§3.3) runs end to end, on a phone, with real AI, without a stumble — twice in a row.

12. Definition of done

The PoC is done when:

- The Daily Brief generates live from real computed signals over the synthetic network, prioritises correctly, connects the Tafo revenue–claims story, and renders with working inline actions.
- The full demo narrative runs end to end on a phone, with real Claude, without a broken screen — reproducibly.
- All twelve facilities are present, alive, and drillable; the network pulse reflects real state.
- Ask answers grounded questions and generates at least a manager message and a facility/board summary.
- My Day, Approvals, and Reports are functional — a brief item can become a task; an approval can be signed; a report can be generated.
- The product is genuinely responsive: mobile-first and a first-class desktop layout, both polished.
- Data is fully Ghana-grounded throughout — cedis, NHIS, MoMo, real towns, real roles — with no generic placeholder content.

Appendix A — Facility roster (detail)

Facility	Region / town	Beds	Pts/mo	Payer mix (N / C+M / Priv)	Notable
Assin Fosu Specialist Hospital	Central / Assin Fosu	60	7,800	65 / 25 / 10	Flagship; the anchor C-site
Asokwa Diagnostic & Specialist	Ashanti / Kumasi	24	5,200	55 / 30 / 15	Stock-out risk (RDT kits)
Kasoa Polyclinic	Central / Kasoa	40	9,400	70 / 25 / 5	High volume; NHIS denial spike
Adansi Community Hospital	Ashanti / Fomena	30	4,100	75 / 22 / 3	Star performer this quarter
Takoradi Harbour Clinic	Western / Takoradi	20	3,600	50 / 35 / 15	Occupational / port health
Tafo Maternity & Child Health	Ashanti / Old Tafo	25	3,900	80 / 18 / 2	Revenue down; claims breakdown
Nima Urban Health Centre	Gt Accra / Nima	18	8,700	68 / 30 / 2	High footfall; long waits
Ho Central Medical Centre	Volta / Ho	35	4,800	72 / 23 / 5	Regional general
Tamale North Clinic	Northern / Tamale	22	4,300	78 / 20 / 2	Attrition + licence expiry
Cape Coast Castle Clinic	Central / Cape Coast	28	4,600	66 / 28 / 6	Theatre under-utilised
Sunyani Diagnostic Hub	Bono / Sunyani	16	2,400	60 / 32 / 8	Newly acquired; ramping
Sekondi Pharmacy & Clinic	Western / Sekondi	12	3,100	45 / 45 / 10	Retail pharmacy + clinic

Payer mix key: N = NHIS, C+M = cash & mobile money, Priv = private insurance (percentages).

Appendix B — KPI definitions

Metric	Definition	Ghana note
Network revenue	Total billed revenue across all facilities for the period.	In GH₵; split across NHIS, cash, MoMo, private.
NHIS outstanding	Approved claims not yet reimbursed by the Scheme.	Often large and slow — a real cash-flow drag.
NHIS denial rate	Denied claims ÷ submitted claims.	Frequently a coding/documentation issue, fixable.
Unbilled (leakage)	Services delivered but never billed.	Silent revenue loss; high-value to surface.
Occupancy	Beds occupied ÷ beds available.	Per facility and network-wide.

Metric	Definition	Ghana note
Patients seen	OPD + admissions for the period.	Seasonal — rises with malaria season.
Payer mix	Share of revenue by payer type.	NHIS / cash & MoMo / private.
Facility margin	Revenue minus operating cost per facility.	Drives portfolio and off-take valuation.
Stock-out risk	Items projected to run out inside reorder lead time.	Reagents, antimalarials, consumables.
Network pulse	Composite, AI-assessed health of the whole network.	One glance: how is the empire right now.

Appendix C — Planted stories (demo material)

These are the narratives baked into the data for the AI to surface. They make the brief and the demo compelling and repeatable.

Facility	What's happening in the data	What the cockpit surfaces
Tafo Maternity	Revenue down ~22% over two weeks; volume flat; NHIS claims recorded but not submitted; ~GH₵ 78k unbilled.	Top brief item: a claims-officer breakdown, not lost demand. Offers "Why?" and "Message manager."
Asokwa Diagnostic	Malaria RDT kit stock burns down; ~5 days left; supplier lead time 7 days.	Watch item: reorder today or testing stops by the weekend. Offers "Approve reorder."
Adansi Community	Best week this quarter: OPD +14%, clean claims, no stock issues.	Positive item: study and replicate to Kasoa and Nima.
Kasoa Polyclinic	NHIS denial rate spiking on a coding issue; high volume amplifies the loss.	Surfaced as a fixable revenue problem with a clear cause.
Tamale North	A key Physician Assistant shows attrition-risk signals; a licence expires soon.	Staff-risk flag with the licence-expiry warning.
Cape Coast	Surgical theatre sitting well below capacity — idle, billable capacity.	Opportunity flag: under-utilised theatre = lost revenue.
Sunyani Hub	Lower numbers — but newly acquired and ramping.	Explicitly framed as expected, not a problem (shows judgement).
Nima Urban	Very high footfall; average wait times climbing.	Operational bottleneck flag.
Approvals	GH₵ 85k ultrasound (Assin Fosu); new Medical Officer (Kasoa); generator servicing (Nima).	Sit in the approvals queue and are named in the brief.

REMEMBER WHY WE'RE BUILDING THIS

The cockpit is the wedge, not the whole. It proves — in sixty seconds, in his hand — that XCreativs can give Gigmann a brain. When he commits to it, the interoperability layer and the full intelligence engine are what he pays to unlock next. Build the brief so well that the rest of the conversation becomes inevitable.